Department of Electrical Engineering
University of Arkansas

ELEG 31204 SYSTEMS AND SIGNALS

Ch. 0 Preface

Dr. Jingxian Wu

wuj@uark.edu

GENERAL INFORMATION

- **Instructor: Dr. Jingxian Wu**
  - Office: Bell 3168
  - Email: wuj@uark.edu
  - Phone: (479) 575-6584

- **Office Hours**
  - Tue. 11:00 AM – 12:00 PM or by appointment

- **Lecture Schedule**
  - Tu. Th. 12:30 PM– 1:45 PM
  - Bell 2273

COURSE MATERIALS

- **Required Course Materials (Free!)**
  - Lecture notes, homework, lab manual
  - Can be downloaded at the course website
    - Main link: https://wuj.hosted.uark.edu/teaching/eleg3124/eleg3124.html
    - Alternative link via Open Education Resource (OER) https://scholarworks.uark.edu/oer/5/
- **Text Books (optional)**
  - Luis F. Chaparro, *Signals and Systems Using Matlab*, Academic Press, 2010. ISBN :978-0-12-374716-7.
- **Required Software**
  - Matlab
  - Free for all UofA students
- **References (optional)**
  - S.S. Soliman, M.D. Srinath, *Continuous and discrete signals and systems*, 2nd Ed., Prentice Hall, 1998.

COURSE INFORMATION

- **Pre-requisite:**
  - Calculus I, II, and III, Differential Equation (co-requisite)
  - Electrical Circuits I
- **This course involves heavy mathematical derivations**
  - It is a Mathematics course.
    - Integration, differentiation, differential equation, etc.
    - Examples with detailed step-by-step derivation will be given during lecture
      - A great opportunity for you to review and practice your Math skills!
  - A large number of examples will be given in class – It's ESSENTIAL for you to repeat all the examples by yourself after class.
  - Homework solutions will be posted after the due dates.
  - You are encouraged to study in groups on homework assignments.
  - Use the office hours of the TA.
- **Teaching format**
  - Slides
  - Examples
  - Exercises
  - Homework assignments (problems, software)
  - Labs

TEST

- **Tests**
  - Three tests. Each covers 1/3 of the semester.
    - No comprehensive test.
  - You can bring a double-sided equation sheet
    - Handwritten
    - Equations only
    - No text description
    - Upload the equation sheet together with your answer sheet (attach the equation sheet to the end of the test).
  - The questions will be very similar to examples in class or homework problems.

TENTATIVE SCHEDULE

- Week 1 (8/18, 8/20): Ch.1 Continuous-Time Signals
- Week 2 (8/25, 8/27): Ch. 1 Continuous-Time Signals
- Week 3 (9/1, 9/3): Ch. 2 Continuous-Time Systems
- Week 4 (9/8, 9/10): Ch. 2 Continuous-Time Systems
- Week 5 (9/15, 9/17): Ch. 2 Continuous-Time Systems
- Week 6 (9/22, 9/24): Ch. 3 Fourier Series
- Week 7 (9/29, 10/1): Ch. 3 Fourier Series (**Test 1 on 10/1**)
- Week 8 (10/6, 10/8): Ch. 3 Fourier Series
- Week 9 (10/13, 10/15): Ch. 4 Fourier Transform
- Week 10 (10/20, 10/22): Ch. 4 Fourier Transform (**Fall break 10/20**)
- Week 11 (10/27, 10/29): Ch. 4 Fourier Transform
- Week 12 (11/3, 11/5): Ch. 5 Laplace Transform (**Test 2 on 11/6**)
- Week 13 (11/10, 11/12): Ch. 5 Laplace Transform
- Week 14 (11/17, 11/19): Ch. 5 Laplace Transform
- Week 15 (11/24, 11/26): Ch. 5 Laplace Transform (**Thanksgiving 11/27**)
- Week 16 (12/1, 12/3): Ch. 6 Discrete-Time System (reading day: 12/4)
- Test 3 in final week (**week of 12/7, date and time TBD**)

GRADING POLICY

- **Grades Percentage**
  - Test 1 22%
  - Test 2 22%
  - Test 3 22%
  - Homework 14%
  - Lab 15%
  - Quiz 5%

- **Grades**
  - A: $90 \leq \text{grade} \leq 100$
  - B: $80 \leq \text{grade} < 90$
  - C: $70 \leq \text{grade} < 80$
  - D: $60 \leq \text{grade} < 70$
  - F: $0 \leq \text{grade} < 60$

GRADING POLICY

- Due dates for homework and lab report will be strictly enforced. Late submission within one week after due will receive a 20% deduction; no credit if submitted one week past due.

- Homework and lab reports should be submitted directly on blackboard

- There will be NO make up for quizzes.

- If for some legitimate reason (sickness, death in the family, etc.), you cannot take an exam on the scheduled day, you must notify the instructor prior to the exam.

ONLINE RESOURCES

- **Course Home Page**
  - https://wuj.hosted.uark.edu/teaching/eleg3124/eleg3124.html
  - All the course related materials, such as slides, homework assignments, lecture notes, homework solutions, links, announcements, etc., will be posted on this website.
  - Please check the webpage regularly (at least once per week) for update.
  - Blackboard

ADDITIONAL ISSUES

- **Academic Honesty**
  - Each University of Arkansas student is required to be familiar with and abide by the University's 'Academic Integrity Policy' which may be found at http://provost.uark.edu/
  - Any kind of activities related to academic dishonesty (copying homework, lab report, code, plagiarism, etc.) will be dealt with.
  - If you are not sure about plagiarism, please contact the instructor.
- **Questions are welcome in my class**
  - You are very welcome to raise any question related to course materials.
  - Please feel free to stop me at any time if you have any question.
  - You can also ask me question via email or during office hours.
- **BE ON TIME!**
- **No texting or web surfing during class.**
- **Have Fun!**